# Truly Automated Formal Specification and Verification with Near-Zero Human Labour

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## ABSTRACT

Formal verification has historically demanded deep expertise in proof assistants and months of dedicated human effort, limiting its adoption to a handful of well-funded teams. We show an example of how LLM-driven automated agentic workflows can shift this cost nearly entirely to compute. Using *Lean Squad*, an open-source workflow that runs as a scheduled GitHub Actions job, we produce a machine-checked formal specification and verification of core safety properties of `raft-rs`—a production Rust implementation of the Raft distributed consensus protocol—with near-zero human labour. The human author required no expertise in Lean, formal methods, or proof engineering; his role was limited to designing the reusable workflow, merging pull requests, and writing two short comments across 34 autonomous agent runs.

The agent autonomously researched the codebase, selected 22 verification targets, designed a six-layer proof architecture (data structures, quorum arithmetic, protocol operations, cross-module composition, Raft safety, and reachability), wrote all specifications, and completed all proofs: 443 theorems across 23 Lean 4 files (stdlib only, no Mathlib), culminating in the end-to-end theorem `raftReachable_safe`—for any cluster state reachable via valid Raft transitions from an empty initial state, no two nodes ever apply different log entries at the same committed index—with zero `sorry`. One formulation bug was caught by the `sorry` discipline; no implementation bugs were found, providing machine-checked evidence of correctness on the verified paths.

The total LLM token cost was approximately $240 (34 runs at ~$7 each using Claude Opus 4.6 pricing at time of publication). The Lean Squad workflow is open-source, installs on any GitHub repository with two commands, and requires only an LLM API token. The proof artefacts and full run history are available at `dsyme/raft-lean-squad`.

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**Note on authorship and voice.** Throughout this paper, "we" generally reflects the perspective of *LeanSquad*, an autonomous AI automation workflow implemented using GitHub Agentic Workflows, with *Copilot CLI* (GitHub) as the underlying AI coding agent using the *Claude Opus 4.6* (Anthropic) LLM. The human author (Don Syme) took a deliberately *hands-off* approach: he designed the automated Lean Squad workflow itself (§3) and merged pull requests, but was not involved in target selection, the six-layer proof architecture, specification writing, or any actual proof engineering. His total intervention over the project's 34 runs amounted to two short GitHub comments (§6.3).

## 1 INTRODUCTION

Distributed consensus algorithms are notoriously subtle. The Raft protocol was designed to be understandable [33], yet its correctness argument spans term management, log matching, leader completeness, and state-machine safety across multiple interacting invariants. Production implementations add further complexity: joint membership changes, in-flight message tracking, flow control, and optimised quorum computation.

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Formal verification of distributed systems is well-established but typically expensive: IronFleet [17] required person-years of Dafny proof engineering; Woos et al. [54] verified a simplified Raft in ~50,000 lines of Coq. The expertise barrier—fluency in a proof assistant, invariant discovery, and multi-month proof campaigns—has limited adoption to a handful of well-funded teams. Our contribution addresses this barrier directly, and is different in three ways:

(1) We target a *production* Raft implementation (`raft-rs`, used in TiKV [3]), not a pedagogical model.
(2) We use Lean 4 *stdlib only*—no Mathlib—providing a lightweight, self-contained verification artefact.
(3) The entire effort is *automated*: every target selection, specification, proof architecture decision, and proof was produced by an LLM-driven FV agent across 34 independent runs, with only two brief human interventions (§6.3). The human author required no Lean expertise, no formal-methods background, and invested near-zero hours of proof-engineering labour.

The top-level theorem (§5) is:

```
theorem raftReachable_safe [DecidableEq E]
    (hd : Nat) (tl : List Nat) (cs : ClusterState E)
    (hvoters : cs.voters = hd :: tl)
    (h : RaftReachable cs) : isClusterSafe cs
```

This is machine-checked by Lean 4 with zero `sorry`.

# 2 BACKGROUND

## 2.1 The Raft Consensus Protocol

Raft replicates a state machine through a distributed log. A cluster of $n$ nodes elects a leader; the leader accepts log entries from clients and replicates them to followers via `AppendEntries` RPCs. An entry is *committed* once a majority quorum of nodes has acknowledged it. Safety requires that no two nodes ever apply different entries at the same log index.

The `raft-rs` codebase (derived from TiKV's raft-rs [3]) implements the Consensus Module in Rust: 52 source files covering quorum arithmetic, log management, flow control, membership change, and configuration validation. It supports joint consensus for membership transitions, a ring-buffer-based in-flight message tracker, and per-node progress tracking.

## 2.2 Lean 4

Lean 4 [8] is a dependently typed proof assistant and programming language. Proofs are written in a tactic language with automation including `omega` (linear arithmetic), `simp` (rewriting), and `decide`/`native_decide` (decidable propositions). We use Lean 4.28.0, stdlib only. All proofs are machine-checked by the Lean kernel; the `sorry` keyword marks unproved obligations.

# 3 THE LEAN SQUAD WORKFLOW

The Lean Squad workflow [1] is a GitHub Agentic Workflow—a declarative specification that orchestrates an LLM coding agent (e.g. Copilot CLI or Codex CLI) as a scheduled GitHub Actions job. It was designed by the human author and open-sourced as a reusable workflow template; it is not specific to Raft or to this repository. Once installed, the workflow runs autonomously on a schedule (every 8 hours in our deployment), with no human in the loop for any individual run.

## 3.1 Design Principles

Four design principles underpin the workflow:

*Zero human involvement necessary.* The agent handles the entire FV effort: research, target selection, proof architecture design, specification writing, and proof engineering. The human need only merge PRs and occasionally nudge the process. This shifts the cost of formal verification from scarce human proof-engineering expertise to LLM compute, dramatically lowering the barrier to entry and making FV a practical option for any codebase—not just those with dedicated verification teams.

*Optimistic.* The agent does not wait for perfect conditions. It starts where the code is and finds what can be verified, even if that means working with approximations or partial proofs.

*Incremental.* Each run contributes one small piece. A spec with `sorry` today becomes a proof next week. A research note today drives a formal spec next month. Persistent memory ensures each run builds on the last.

*Findings-first.* A counterexample is a success, not a failure. Finding a bug through formal verification is exactly the point.

*Transparent.* Every PR, issue, and comment includes an AI disclosure. Specs document their approximations. Proofs document what is still open. THe history of every run is public, and the full history of the specifications and proofs is available in the repository for the humand and the agent to examing as needed.

## 3.2 Phase-Weighted Task Selection

Each run begins with a deterministic pre-step that inspects the repository's FV artefacts (Lean files, spec documents, open issues and PRs) and computes a phase-weighted probability distribution over ten tasks. Two tasks are drawn and communicated to the agent; the agent confirms them against its persistent memory and executes them. A mandatory final task (status-issue update) always runs.

The ten tasks, in order of the typical progression, are:

(1) **Research & Target Identification.** Survey the codebase to identify FV-amenable functions, data structures, and algorithms. Document expected benefit, spec size, proof tractability, and recommended approach. Produces `RESEARCH.md` and `TARGETS.md`. If a critique document exists, incorporate its feedback to adjust priorities.
(2) **Informal Spec Extraction.** For the highest-priority unstarted target, extract a precise informal specification by reading the code and inferring the design intention—preconditions, postconditions, invariants, edge cases, concrete examples, and inferred intent not explicit in the code. Flags ambiguities for maintainer review.
(3) **Formal Spec Writing (Lean 4).** Write Lean 4 type definitions, function stubs, and key `theorem` declarations with `sorry` as proof bodies. Must pass `lake build` before opening a PR.
(4) **Implementation Extraction.** Translate the relevant Rust implementation into Lean 4 functional definitions so propositions can be formally connected to actual code. For imperative code, create a pure functional model and document the abstraction.
(5) **Proof Assistance.** Attempt to prove stated propositions using Lean 4 tactics. When a proof obligation cannot be closed, investigate whether the spec or the implementation is wrong. If a counterexample is found, file a bug issue.
(6) **Correspondence Review.** Review how faithfully each Lean model corresponds to the Rust source. Assess correspondence level (exact, abstraction, approximation, or mismatch) and document all divergences.
(7) **Proof Utility Critique.** Honestly assess whether the FV work is useful—are the proved properties meaningful, at the right level of abstraction, and likely to catch real bugs? Identify highest-value gaps.
(8) **CI Automation.** Set up and maintain `lean-ci.yml` to automatically verify Lean proofs on every PR and push.
(9) **Project Report.** Create and maintain `REPORT.md`—a comprehensive report with mermaid diagrams summarising the entire FV effort.
(10) **Conference Paper.** Write and maintain a LaTeX paper under `formal-verification/paper/`, compile to PDF, and commit to the repository.

The weighting scheme adapts automatically to the current state of the repository. When no FV work exists, Task 1 (Research) dominates. Once research is done, Task 2 rises. As informal specs accumulate, Task 3 gains weight, and so on up through proofs and beyond to reflective tasks (critique, correspondence review, paper writing). Table 1 illustrates the weight progression.

| Task | Name | Weight (early) | Weight (late) |
|---|---|---|---|
| 1 | Research | 10.0 | 2.0 |
| 2 | Informal Spec | 8.0 | 2.0 |
| 3 | Formal Spec | 8.0 | 2.0 |
| 4 | Implementation | 6.0 | 3.0 |
| 5 | Proof Assistance | 6.0 | 5.0 |
| 6 | Correspondence | 10.0 | 5.0 |
| 7 | Critique | 10.0 | 5.0 |
| 8 | CI | 12.0 | 1.0 |
| 9 | Report | 8.0 | 3.0 |
| 10 | Paper | 8.0 | 3.0 |

**Table 1: Phase-weighted task selection. "Early" weights apply when the task's prerequisites are met but the task has not yet been completed; "late" weights apply once the artefact exists and is being maintained.**

## 3.3 Persistent Memory and Cumulative Progress

A critical design element is persistent repo-memory. At the start of every run, the agent reads its memory to recover the identified FV targets, their current phases (1–5), key choices, open questions, bugs found, and ideas to try. At the end of every run, the agent updates memory with new discoveries, proved theorems, and remaining obligations.

Memory may be stale (referencing merged PRs or closed issues), so the agent is instructed to verify references against the current repository state. This mechanism enables cumulative progress across independent runs: the six-layer architecture, the choice of targets, and the proof strategy all emerged organically from the agent's interaction with its own memory and the growing codebase, not from human direction.

## 3.4 Hard Requirements

The workflow enforces several hard requirements to maintain integrity:

- The Lean toolchain **must** be successfully installed before any `.lean` files are written or submitted. If installation fails, Tasks 3–5 are skipped entirely for that run.
- Every PR that includes `.lean` files must include a verification status block: either "`lake build` passed" or an explicit disclosure that proofs are unverified.
- The agent must never claim proofs are verified unless `lake build` actually passed.

- All PRs, issues, and comments must include an AI transparency disclosure.

### 3.5 Artefacts Produced

The workflow produces a rich set of artefacts beyond Lean proofs: `RESEARCH.md` (FV survey and rationale), `TARGETS.md` (prioritised target list with phase status), per-target informal specs, `CORRESPONDENCE.md` (Lean-to-Rust mapping), `CRITIQUE.md` (proof utility assessment), `REPORT.md` (comprehensive project report), and the conference paper itself. A continuously-updated GitHub issue serves as a rolling dashboard showing the status of every target.

### 3.6 Installation and Usage

The Lean Squad workflow is open-source and can be installed on any GitHub repository with two commands. A token for an LLM coding agent is required; at the time of writing, Copilot CLI (GitHub), Codex CLI (OpenAI), and Claude Code (Anthropic) are supported as underlying agents.

```
# Install the GitHub Agentic Workflows CLI extension
gh extension install github/gh-aw

# Add Lean Squad to the current repository
gh aw add-wizard githubnext/agentics/lean-squad
```

The `add-wizard` command walks the user through configuration: selecting the schedule (default: every 8 hours), choosing the agent provider, and storing the API token as a GitHub Actions secret. Once committed, the workflow runs autonomously on the configured schedule with no further setup.

With just these two commands, any repository can start an automated formal verification effort with zero human specification or proof-engineering expertise required. Whether that is useful or not depends on the codebase and the agent's performance, but the barrier to entry is minimal. Getting started with formal verification of real software has never been easier.

## 4 METHODOLOGY

### 4.1 Target Selection

In its very first run (PR #93), the agent surveyed the `raft-rs` codebase via Task 1 (Research) and identified an initial set of **10 FV targets**, prioritised by tractability: `limit_size`, `Config::validate`, `vote_result`, `committed_index`, `find_conflict`, `maybe_append`, joint-config variants, `inflights`, and `progress`. That same run also produced the first informal specification (`limit_size`). The criteria were: pure functions with clear pre/postconditions, data-structure invariants, and proof tractability. No human input was involved in selecting or prioritising targets.

Notably, the first run's research plan proposed Lean 4 + Mathlib and flagged the Aeneas extraction toolchain as a possible route; both choices were later revised by the agent itself—switching to stdlib-only and dropping Aeneas—as it gained experience with the proof artefacts across subsequent runs.

Over later iterations the agent expanded the target list to **22 targets** and organised them into six layers (Table 2). Twelve additional targets were added as the agent discovered cross-module composition opportunities (`SafetyComposition`, `CrossModuleComposition`), protocol-level properties (`RaftSafety`, `RaftProtocol`), the reachability model (`RaftTrace`), and further quorum lemmas (`HasQuorum`, `TallyVotes`, `QuorumRecentlyActive`, `IsUpToDate`, `LogUnstable`). This layered architecture emerged from the agent's own analysis of module dependencies—it was not prescribed by the human author. Layers build on each other: Layer 2 (Quorum Arithmetic) uses types from Layer 1; Layer 4 (Composition) spans Layer 2 and Layer 3; Layers 5–6 depend on all preceding layers.

| Layer | Description | Files | Theorems |
|---|---|---|---|
| 1 | Data Structures | 5 | ≈144 |
| 2 | Quorum Arithmetic | 7 | ≈116 |
| 3 | Protocol Operations | 4 | ≈63 |
| 4 | Cross-Module Composition | 3 | 26 |
| 5 | Raft Safety | 2 | 24 |
| 6 | Reachability | 1 | 3 |
| | **Total** | **22** | **443+** |

**Table 2: Proof layers and theorem counts.**

### 4.2 Specification Strategy

Each target was formalised in two phases. First, an *informal specification* was extracted by reading the Rust source, its tests, and documentation comments. This produced a prose document identifying preconditions, postconditions, invariants, edge cases, and open questions.

Second, the informal spec drove a *Lean 4 formalisation*: type definitions mirroring the Rust types (with deliberate approximations documented in CORRESPONDENCE.md), function stubs, and theorem statements. Proofs were then attempted incrementally, from simplest (structural induction, `omega`) to most complex (multi-lemma composition).

### 4.3 Modelling Choices

The Lean model deliberately abstracts several Rust features:

*Type abstraction.* `u64` voter IDs and index values are modelled as `Nat`. This eliminates overflow concerns (safe: no

practical Raft deployment approaches $2^{64}$) and enables `omega` arithmetic.

*Collection abstraction.* `HashSet<u64>` voter sets are modelled as `List Nat`. Theorems that require uniqueness carry an explicit `voters.Nodup` hypothesis; those that do not hold for arbitrary lists.

*Log abstraction.* The per-node log is modelled as `VoterLogs E := Nat → Nat → Option E`—a pure function from (voter, index) to optional entry. This abstracts over the Rust `Vec`-based log structure while preserving the read semantics.

*Protocol abstraction.* Concrete Raft message types (`AppendEntries`, `RequestVote`), term management, and leader election are not modelled. They are instead captured by the abstract `RaftReachable` step hypotheses (§6.1).

*Size semantics.* For `limit_size`, protobuf message size is abstracted as an arbitrary `size : α → Nat` function. Proofs hold for any such function, making them strictly more general than the Rust implementation.

## 4.4 Proof Approach

We used the following tactic strategy, from simplest to most complex:

- `omega`: linear integer/natural arithmetic (most common)
- `simp`/`simp only`: definitional unfolding and rewriting
- `by_cases`: boolean and decidable case splits
- `induction`/`cases`: structural recursion on lists and inductives
- `exact`/`apply`/`refine`: explicit term construction
- `funext`: function extensionality

We used no `native_decide`, no axioms beyond Lean's kernel, and no `sorry` in the final proof state.

# 5 RESULTS

## 5.1 Layer-by-Layer Proof Inventory

*Layer 1: Data Structures.* Five files prove correctness of individual Raft data structures in isolation.

`LimitSize.lean` (17 theorems) formalises the Rust function `util::limit_size`, which truncates a vector of protobuf messages to stay within a byte budget. Key results: *prefix correctness* (the result is a prefix of the input), *non-emptiness* ($\geq 1$ entry always preserved), and *maximality* (the longest prefix within budget).

`Inflights.lean` (49 theorems) is the largest single file. It formalises the ring-buffer-based in-flight message tracker (`inflights.rs`), proving the abstract/concrete bridge: the efficient ring buffer and a simple list model compute identical `free_to` results.

`Progress.lean` (31 theorems) formalises the per-node progress tracker state machine, proving the well-foundedness invariant `matched + 1 ≤ next_idx` is preserved across all transitions.

*Layer 2: Quorum Arithmetic.* Seven files prove the mathematical foundations of Raft safety. The cornerstone is `quorum_intersection_mem` (HQ20, `HasQuorum.lean`):

```
theorem quorum_intersection_mem
  (voters : List Nat) (hne : voters     [])
  (a b : Nat     Bool)
  (ha : hasQuorum voters a = true)
  (hb : hasQuorum voters b = true) :
      v, v     voters     a v = true     b v = true
```

This states: any two majority quorums over the same voter list share at least one concrete member. This is the quorum-intersection property that makes Raft (and Paxos) safe. The proof proceeds by an arithmetic argument using inclusion-exclusion and `omega`.

`CommittedIndex.lean` (17 theorems) formalises the *sort-then-index* algorithm that computes the committed index from a list of per-voter match indices, proving both safety (result is attested by a majority) and maximality (result is the largest such index).

*Layer 3: Protocol Operations.* Four files prove correctness of core Raft log operations. `MaybeAppend.lean` (18 theorems) gives a complete post-condition for `maybe_append`: theorem MA13 proves the prefix is untouched and MA14 proves the suffix is correctly applied. Together they provide a behavioural specification of AppendEntries log modification.

`FindConflict.lean` (12 theorems) proves that the function returns zero iff all entries match (FC11), and returns the first index of mismatch otherwise.

*Layer 4: Cross-Module Composition.* Three files prove properties spanning multiple independent modules. `SafetyComposition.lean` (9 theorems) bridges quorum arithmetic and committed index computation: theorem SC4 states that if two voters have committed different entries at the same index, they must share a witness voter in their respective quorums—but the same voter cannot have voted for two different entries, giving a contradiction.

`CrossModuleComposition.lean` (7 theorems) proves that `maybe_append` is safe with respect to quorum commitments (CMC3): it never advances a committed index beyond what the current quorum has certified.

*Layer 5: Raft Safety.* `RaftSafety.lean` (14 theorems) lifts quorum-level safety to the log-entry level. The central theorem is RSS1:

```
theorem raft_state_machine_safety
  [DecidableEq E]
  (voters : List Nat) (hne : voters     [])
  (logs : VoterLogs E) (k : Nat)
  ( e    e   : E)
```

```
  (h1 : isQuorumCommitted voters logs k  e  )
  (h2 : isQuorumCommitted voters logs k  e  )
  :  e   =  e
```

If two entries are simultaneously quorum-committed at the same index, they must be identical. The proof uses `quorum_intersection_mem` (HQ20) to obtain a shared witness voter and conclude that both entries equal the witness's log entry.

RSS8 (`raft_end_to_end_safety_full`) provides the conditional end-to-end result: given the `CommitCertInvariant`—every applied entry is quorum-certified—the cluster is safe.

`RaftProtocol.lean` (10 theorems) proves that `AppendEntries` preserves the Log Matching Invariant (RP6) and the No-Rollback property (RP8), formalising the Raft protocol transition rules at the log level.

```
inductive RaftReachable [DecidableEq E] :
  ClusterState E     Prop where
| init (voters : List Nat) :
    RaftReachable (initialCluster voters)
| step {cs cs' : ClusterState E}
    (hreach : RaftReachable cs)
    (hvoters : cs.voters = cs'.voters)
    (v : Nat)
    (hlogs' :     w, w     v
      cs'.logs w = cs.logs w)
    (hno_overwrite :     k,
      cs.committed v     k
      cs'.logs v k = cs.logs v k)
    (hqc_preserved :     k e,
      isQuorumCommitted
        cs.voters cs.logs k e
          w, cs'.logs w k = cs.logs w k)
    (hcommitted_mono :     w,
      cs'.committed w     cs.committed w)
    (hnew_cert :     w k e,
      cs'.committed w     k
      cs.committed w < k
      cs'.logs w k = some e
      isQuorumCommitted
        cs'.voters cs'.logs k e) :
    RaftReachable cs'
```

**Figure 1: The `RaftReachable` inductive protocol model.**

*Layer 6: Reachability.* `RaftTrace.lean` (3 theorems) closes the gap between the conditional safety result (RSS8) and an unconditional statement. It defines the `RaftReachable` inductive predicate (Figure 1) and proves by structural induction that every reachable state satisfies `CommitCertInvariant` (RT1), from which RSS8 yields the top-level theorem RT2:

```
theorem raftReachable_safe [DecidableEq E]
    (hd : Nat) (tl : List Nat) (cs : ClusterState E)
    (hvoters : cs.voters = hd :: tl)
    (h : RaftReachable cs) : isClusterSafe cs :=
  raft_end_to_end_safety_full hd tl cs hvoters
    (raftReachable_cci cs h)
```

The two-line proof follows immediately from RT1 and RSS8.

## 5.2 Proof Statistics

Table 3 summarises the full file inventory. Total: 443+ theorems, 23 files, 0 `sorry`, 7,440 lines of Lean.

| File | Thms | Key result |
|---|---|---|
| LimitSize | 17 | Prefix, maximality |
| ConfigValidate | 10 | 8-constraint predicate equiv |
| MajorityVote | 21 | Won/Lost/Pending characterisation |
| JointVote | 14 | Joint-config degeneracy |
| CommittedIndex | 17 | Sort-index safety + maximality |
| FindConflict | 12 | First mismatch; zero ⇔ all-match |
| JointCommittedIndex | 10 | $\min(ci_{in}, ci_{out})$ |
| MaybeAppend | 18 | Prefix preserved; suffix applied |
| Inflights | 49 | Ring-buffer abstract/concrete bridge |
| Progress | 31 | State-machine WF invariant |
| IsUpToDate | 18 | Lex order, total preorder |
| LogUnstable | 37 | Snapshot/entries consistency |
| TallyVotes | 18 | Partition, bounds, no-double-quorum |
| HasQuorum | 22 | Quorum intersection (HQ14, HQ20) |
| QuorumRecentlyActive | 15 | Active-quorum detection |
| SafetyComposition | 9 | SC4: two CIs share witness |
| JointTally | 14 | Joint-config tally correctness |
| JointSafetyComposition | 10 | JSC7: witnesses in both halves |
| CrossModuleComposition | 7 | CMC3: maybe_append bounded |
| RaftSafety | 14 | RSS1–RSS8: cluster safety |
| RaftProtocol | 10 | LMI, NRI preserved by AppendEntries |
| RaftTrace | 3 | RT1, RT2: unconditional safety |
| Basic | – | Shared definitions |
| **Total** | **443+** | **0 sorry** |

**Table 3: Full file inventory (all at Phase 5).**

## 5.3 Bugs and Findings

*No implementation bugs found.* All 443+ theorems are consistent with the Rust implementation. This is itself a meaningful positive finding: the quorum arithmetic, log operations,

config validation, flow control, and state-machine invariants are all consistent with their formal specifications on the verified paths.

*Formulation bug caught and corrected.* Early versions of `log_matching_property` (RSS3) and `raft_committed_no_rollback` (RSS4) claimed properties for *arbitrary* log states—with no hypotheses about how the logs were constructed. These statements are provably false (a log with random contents satisfies no log-matching property). The `sorry` mechanism acted as a "needs review" marker: since `sorry` was required but no proof could be found, the error was caught before it entered the proof base.

The corrected versions carry explicit hypotheses: `LogMatchingInvariantFor` and `NoRollbackInvariantFor`, which are exactly the invariants preserved by valid Raft transitions. Both were subsequently proved (run r130).

# 6 DISCUSSION

## 6.1 Proof Architecture and the Residual Gap

The `RaftReachable` model (Figure 1) is the honest accounting of what has and has not been proved. The five `step` hypotheses (`hlogs'`, `hno_overwrite`, `hqc_preserved`, `hcommitted_mono`, `hnew_cert`) are proof-engineering artefacts: they precisely state what a protocol transition must satisfy to preserve `CommitCertInvariant`.

The residual gap is that these hypotheses are *assumed* rather than derived from concrete Raft messages. Closing the gap requires: (a) formalising election safety (an elected leader has all quorum-certified entries), (b) the commit rule (`leaderCommit` reflects a genuine quorum acknowledgement), and (c) term monotonicity. This is standard Raft proof-engineering and would be a natural extension of the current work.

Concretely, `hno_overwrite` corresponds to the Rust panic guard:

```
if conflict != 0
   && conflict <= self.committed {
  panic!("entry␣{}␣conflict
␣␣␣␣with␣committed␣log", conflict);
}
```

This panic is proved to be unreachable in a correct Raft execution by `hno_overwrite`. Theorem RP8 (`raft_transitions_no_rollback`) proves that `maybe_append` satisfies `hno_overwrite` given the appropriate preconditions.

## 6.2 Proof Utility Assessment

The six-layer architecture ensures that every theorem in upper layers has a genuine dependency on lower layers. In particular:

- HQ20 (`quorum_intersection_mem`) is the *only* mathematical fact about quorums used in the RSS1 proof. If it were wrong, RSS1 would fail. Its proof is non-trivial (inclusion-exclusion + arithmetic), giving meaningful assurance.
- MA13/MA14 (`MaybeAppend`) give a complete behavioural characterisation of AppendEntries log modification—the most frequent Raft protocol operation. A bug in `maybe_append` would cause one of these to fail.
- SC4 (`SafetyComposition`) is the first theorem that cannot be stated within a single module; it requires both quorum arithmetic (Layer 2) and committed-index computation (Layer 2) to interact.

The main limitation is the abstracted protocol layer: theorems about terms, leader election, and message routing are not included. `raftReachable_safe` is unconditional *within the RaftReachable model*; fully unconditional safety with respect to the Rust implementation would require the extensions in §6.1.

## 6.3 Automation and Human Intervention

The entire FV effort was conducted by the *Lean Squad* automated agent, an LLM-driven agent that runs as a GitHub Actions workflow (§3). Each run: reads persistent memory, draws two tasks from a phase-weighted schedule, installs the Lean toolchain, executes the tasks (writing or modifying Lean files, specs, or documentation), runs `lake build`, and creates a pull request with verified code. A final status-issue update always runs.

The human author's role was deliberately hands-off: he designed the Lean Squad workflow itself (§3), installed it on the repository, and reviewed merged PRs. Over the entire 34-run project, his total intervention consisted of exactly two GitHub comments:

*Intervention 1 (Issue #59, run ≈20).* After 18 targets were fully proved at the data-structure and log-operation level, the human wrote:

> `/repo-assist continue on this work, add more planning as an issue here or do more work via PRs or both`

This triggered a companion agent (*Repo Assist*) to create a planning epic (Issue #59) that surveyed the state of progress and recommended next steps—including the Tier 2 safety invariants (term monotonicity, vote commitment, election safety) that would later inform Layers 5–6.

*Intervention 2 (Issue #94, run ≈28).* After the agent had proved 370 theorems but had not yet attempted state-machine-level safety, the human wrote:

```
/lean-squad Address these concerns from
critique.md, plan out rest of work, write
theorems even if unproven, get the project
on Trajectory to completion
```

with a one-line note: "The main remaining gap is the state-machine level: no end-to-end Raft safety theorem exists yet." This triggered the agent to create `RaftSafety.lean` and `CrossModuleComposition.lean` (run r128), adding 25 new theorems including the first end-to-end cluster safety results. Subsequent runs (r129–r134) completed the remaining sorry-guarded theorems and produced the final `raftReachable_safe`.

*What is remarkable.* is not that two interventions occurred, but what did *not* require human intervention: the choice of all 22 FV targets, the six-layer proof architecture, all 22 informal specifications, all Lean formalisations, all 443+ proofs, the correspondence review, the proof utility critique, the project report, and even an initial draft of this paper were produced autonomously. The human never selected a target, wrote a specification, suggested a proof strategy, or debugged a failing proof.

Across 34 runs spanning ≈6 weeks, the agent produced:

- 22 informal specification documents
- 23 Lean 4 source files (7,440 lines)
- 443+ machine-checked theorems
- 1 formulation bug caught and corrected
- 0 implementation bugs

The key insight enabling this level of autonomy is the workflow's phase-weighted task selection (§3.2): the agent always has a well-defined next action, and persistent memory ensures each run builds on a growing base of named lemmas and established infrastructure.

## 6.4 Lean Stdlib vs. Mathlib

We deliberately chose Lean 4 stdlib only, without Mathlib. This keeps the proof artefact self-contained and avoids version-pinning issues with Mathlib's rapid development. The cost is the absence of mature list-processing lemmas and richer automation.

In practice, most proofs required only `omega`, `simp`, `by_cases`, and structural induction—all available in stdlib. The only Lean 4.28 stdlib pitfalls encountered were: `lemma` not being a top-level keyword (use `theorem` instead), `split_ifs` and `push_neg` not being available (use `by_cases` and `omega`), and `le_rfl` not existing (use `Nat.le_refl _`).

## 6.5 Lessons Learned

*Phase-weighted task selection enables autonomy.* The workflow's ten-task structure with adaptive weights means the agent always has a well-defined next action. The agent does not need a human to tell it what to do next—the weight computation, combined with persistent memory, provides sufficient direction. The six-layer proof architecture emerged organically from this process rather than being prescribed.

*Persistent memory is essential for cumulative progress.* An LLM agent can prove theorems in a new module if all dependencies are already available as named lemmas. Persistent memory ensures the agent knows which lemmas exist, which targets are complete, and what strategies have worked or failed in prior runs.

*The* `sorry` *discipline catches formulation bugs.* Proving with `sorry` as a placeholder and then trying to remove it is a reliable method for finding specification errors. Both RSS3 and RSS4 were caught this way.

*One module per run, not the whole codebase.* Each run focused on a single target. This made the Lean context small enough for the LLM to reason about correctly, and produced clean, reviewable pull requests.

*Formal models must be honest about what they omit.* The CORRESPONDENCE.md document records all known divergences between Lean models and Rust source. Making these explicit—rather than claiming full correspondence—is essential for correctly assessing proof value.

*Negative results have value.* Finding no implementation bugs in 443 theorem-checks is itself evidence of implementation quality. The absence of counterexamples in the verified paths raises confidence, even without a fully concrete model.

## 6.6 Cost

Each of the 34 Lean Squad runs consumed approximately $7 of LLM token cost at Claude Opus 4.6 pricing (at time of publication), for a total of approximately $240. This covers the full pipeline: reading persistent memory, researching the codebase, selecting tasks, installing the Lean toolchain, writing or modifying Lean files, running `lake build`, and creating pull requests. The GitHub Actions compute time (Linux runners) is negligible relative to token cost.

For comparison, IronFleet [17] required person-years of expert proof engineering in Dafny; Woos et al. [54] reported ~50,000 lines of Coq proof. Even valuing human proof-engineer time conservatively, the dollar cost of comparable manual efforts exceeds the LLM cost by orders of magnitude. The trade-off is clear: LLM-driven FV substitutes cheap, parallelisable compute for scarce, expensive human expertise. As model prices continue to fall, the economics will only improve.

### 6.7 Training Data Overlap

It is virtually certain that Claude Opus 4.6 was trained on published papers, source code, and proof artefacts related to the formal verification of Raft—including Ongaro's dissertation [32], the Verdi/VRaft Coq development [54], TLA+ specifications, and the `raft-rs` source code itself. The degree to which this prior exposure influenced the agent's target selection, proof architecture, and tactic choices cannot easily be measured.

We note two mitigating factors. First, the proofs are machine-checked by Lean's kernel: regardless of how the agent arrived at a proof term, its correctness is independently verified. Training-data influence cannot produce a false positive. Second, several architectural decisions—stdlib-only (no Mathlib), the six-layer decomposition, the `RaftReachable` abstract model—have no direct precedent in the prior Raft verification literature, suggesting that the agent did more than reproduce memorised material. Nevertheless, this overlap should be borne in mind when assessing the generalisability of our results to domains with less existing formal-verification literature.

## 7 RELATED WORK

We situate our contribution with respect to six threads of prior work.

### 7.1 Formal Verification of Raft

The closest predecessor is Woos et al. [54], who gave the first machine-checked proof of state-machine safety for Raft in Coq using the Verdi framework [53]. Their proof discovered 90 system invariants and extracted a runnable OCaml implementation. A key methodological contribution was *planning for change*: factoring proofs so that strengthening one invariant does not cascade failures throughout the development. Our work differs in three respects: (i) we target a *production* Rust implementation (`raft-rs`) rather than an extracted model; (ii) we use Lean 4 with stdlib only; and (iii) the entire effort is LLM-automated. Our layered architecture echoes their information-hiding methodology, but arises from the need to give the LLM agent a well-defined proof frontier rather than from human proof-maintenance concerns.

Subsequent Raft verification efforts have used diverse formalisms. Evrard [13] modelled Raft in LNT for model-checking with CADP. Bao et al. [4] and Ishibashi and Ogata [22] model-checked Raft leader election in NuSMV and Maude respectively. Sanabria et al. [42] verified Raft leader election safety using the Athena proof system. Bora et al. [7] modelled Raft in mCRL2, enabling behavioural equivalence checking. Xiong et al. [56] formalised split, merge, and membership changes for Raft (ReCraft), complementing our focus on log safety. None of these target a production implementation or produce end-to-end safety proofs at the level of `raftReachable_safe`.

Ongaro's TLA+ specification [32] provides a complementary model-checking approach. TLA+ can exhaustively check finite instances but does not produce machine-checked proofs over unbounded state spaces. Our Lean proofs hold for arbitrary cluster sizes and unbounded execution traces, within the `RaftReachable` model.

### 7.2 Verified Distributed Systems

IronFleet [17] verified Paxos-based replication and a distributed key-value store in Dafny with TLA-style temporal reasoning. The Disel framework [44] introduced dependent separation protocols for modular distributed-system verification in Coq. Sprenger et al. [46] proposed Igloo, combining compositional refinement with separation logic for distributed-system verification. Pîrlea and Sergey [35] mechanised blockchain consensus in Coq, demonstrating that protocol-level safety proofs are tractable in proof assistants.

More recently, Honoré et al. developed a series of verified distributed-object frameworks: Adore [21] for certified reconfiguration, ADO [19] for fault-aware compositional verification, and AdoB [20] for bridging benign and Byzantine consensus. Qiu et al. verified linearisable Byzantine distributed objects (LiDO [37]) and extended this to DAG-based consensus (LiDO-DAG [38]). Kim et al. [26] presented VeriRT, an end-to-end verification framework for real-time distributed systems. Lewchenko et al. [30] verified bolt-on strong consistency specifications. Zhao et al. [62] developed compositional verification of composite Byzantine protocols. Zhang et al. [60] studied inductive invariants for distributed protocols with a focus on proof ergonomics.

Two recent systems-level efforts are particularly relevant. AutoMan [61] combines automatic code generation with manual optimisations for verified distributed-system development, achieving strong guarantees on production-grade systems. Veil [34] provides a framework for automated and interactive verification of transition systems, offering an alternative path to the reachability-style reasoning in our Layer 6.

Blockchain consensus verification has also grown rapidly: Thomsen and Spitters [49] formalised Nakamoto-style proof of stake in Coq; Rashid et al. [39] verified safety properties of Ethereum 2.0 Beacon Chain epoch processing; Schultz et al. [43] verified a distributed dynamic reconfiguration protocol.

### 7.3 Formal Verification of Rust

Verifying Rust programs is an active area with several complementary approaches. RustBelt [25] established the semantic foundations of Rust's type system safety guarantees in

Coq using Iris separation logic. Prusti [2] provides automated deductive verification for Rust via Viper, targeting ownership and memory safety. Creusot [10] translates Rust to WhyML for verification with Why3, handling borrows and ownership. Verus [28] embeds verification conditions directly in Rust using linear ghost types and SMT solving.

Our approach is orthogonal: rather than verifying the Rust code directly, we build a pure functional Lean model of the relevant algorithms and verify properties of that model, documenting the correspondence to Rust in `CORRESPONDENCE.md`. This is a deliberate trade-off: we lose mechanical connection to the Rust source (which Aeneas [18] could provide) but gain the full power of Lean's dependent types and tactic framework for protocol-level reasoning that SMT-based tools struggle with.

Bílý et al. [6] proposed a refinement methodology for distributed Rust programs that could, in principle, bridge our abstract Lean models with the concrete `raft-rs` code. Combining their refinement approach with our verified safety properties is a promising direction for future work.

## 7.4 LLM-Based Theorem Proving

Recent years have seen rapid progress in applying LLMs to formal theorem proving. LeanDojo [58] introduced a benchmark and retrieval-augmented approach for Lean proving, where an LLM retrieves relevant premises from Mathlib before generating tactic steps. LEGO-Prover [52] grows a reusable library of verified lemmas across proof attempts, conceptually similar to our persistent memory mechanism. Baldur [14] demonstrated whole-proof generation and repair with LLMs for Isabelle/HOL. Draft-Sketch-Prove [23] uses informal proofs as sketches that guide formal proof search. Copra [48] employs an LLM agent with backtracking for Lean theorem proving. Polu and Sutskever [36] showed that curriculum learning over formal mathematics enables LLMs to solve increasingly difficult problems. Song et al. [45] explored LLMs as interactive copilots for Lean users.

Bayazıt et al. [5] conducted a case study on the effectiveness of LLMs for verification with proof assistants, finding that LLMs can discharge simple lemmas but struggle with multi-step proofs requiring creative insight. Our experience *diverges*: the Lean Squad agent completed all 443+ theorems including multi-step compositional proofs (Layers 4–6) that require chaining lemmas across modules. We attribute this difference to four factors: (i) the workflow's phase-weighted task selection (§3) decomposes the problem into tractable increments, each with a well-defined interface to the next; (ii) persistent memory across runs enables cumulative progress where each run builds on a growing base of named lemmas and established infrastructure; (iii) the remarkable improvement in raw LLM reasoning power in the intervening period—frontier models in 2025–2026 are substantially more capable at multi-step logical reasoning than those available during the Bayazıt et al. study; and (iv) the planning and tool-use capabilities of modern coding agents (Copilot CLI, Codex CLI), which provide structured multi-step reasoning, file navigation, and iterative error correction that go well beyond single-prompt LLM interactions.

These systems differ from our approach in a fundamental way: they target *benchmark-style* theorem proving, where the goal is to close a given lemma statement in an existing formalisation. The human provides the theorem statement and the library context; the LLM provides the proof. In contrast, *Lean Squad operates at the project level*: it selects what to verify, writes the specifications, builds the library from scratch, and proves the theorems—all within a production codebase that has no prior formalisation.

The closest analogue to our persistent-memory mechanism is LEGO-Prover's growing library [52], but LEGO-Prover operates within a single proof session, whereas Lean Squad accumulates knowledge across 34 independent GitHub Actions runs spanning weeks. AlphaProof [50] demonstrated that reinforcement learning over formal proofs can achieve competition-level mathematical reasoning, but targets Olympiad geometry rather than software verification, and requires massive compute for training.

Yao et al. [59] demonstrated mostly-automated verification of liveness properties using ranking functions, and Gopinathan et al. [15] developed mostly-automated proof repair for verified libraries—both pointing toward increasing automation in formal verification. Díaz et al. [11] introduced incremental certified programming, which shares our goal of making cumulative progress on large proof developments.

## 7.5 Autonomous AI Agents for Software Engineering

SWE-bench [24] established a benchmark for evaluating LLM agents on real-world GitHub issues. SWE-agent [57] demonstrated that purpose-built agent-computer interfaces improve automated issue resolution. Agentless [55] showed that simpler, non-agentic approaches can also resolve a significant fraction of issues.

These agents operate in a *reactive* mode: given a bug report or feature request, they produce a patch. Lean Squad operates in a *proactive, scheduled* mode: it runs every 8 hours regardless of external triggers, inspects the repository state, and advances the project. This distinction is important because formal verification is not naturally triggered by external events—it is a long-running engineering effort that requires sustained, cumulative progress.

The phase-weighted task selection mechanism (§3.2) can be viewed as a form of *automated project management*: the

workflow maintains a notion of project phase, allocates effort accordingly, and adapts as the project matures. This is, to our knowledge, novel among LLM-agent architectures.

### 7.6 Proof Engineering and Large-Scale Verification

Ringer et al. [40] surveyed the landscape of formally verified software engineering, identifying proof maintenance as a central challenge—the same challenge that motivated the "planning for change" methodology of Woos et al. [54]. Dejon et al. [9] provided a systematic guide to proof engineering, and Luan et al. [31] studied compatibility issues across theorem prover versions, a practical concern we sidestep by using Lean 4 stdlib only. The proof-repair line of work by Ringer et al. [41] and Viola et al. [51] addresses proof maintenance across type equivalences, complementing our approach of isolating change behind layer boundaries.

Fiat-Crypto [12] demonstrated automatic synthesis of verified cryptographic arithmetic. The Aeneas toolchain [18, 47] enables mechanical Lean extraction from safe Rust; we note its applicability but do not use it in this work.

To calibrate expectations, we compare the scale of our verification with landmark projects. seL4 [27] verified a production OS microkernel: 8,700 lines of C, ∼200,000 lines of Isabelle/HOL proof, 11 person-years. CertiKOS [16] verified a concurrent OS kernel with 6,500 lines of C and ∼100,000 lines of Coq. CompCert [29] verified an optimising C compiler over ∼100,000 lines of Coq. IronFleet [17] verified Paxos replication with ∼42,000 lines of Dafny proof. Woos et al. [54] verified simplified Raft with ∼50,000 lines of Coq across 90 invariants.

Our 7,440 lines of Lean and 443 theorems are modest by comparison. However, the relevant metric is not raw scale but *effort*: those projects required person-years of expert proof engineering, while ours required zero human proof engineering and approximately 34 LLM agent runs (each ∼15–30 minutes of wall-clock time) plus two one-line human comments. If the approach scales—and the successful progression from Layer 1 through Layer 6 suggests it can—the implications for the cost of formal verification are significant.

Our contribution is, to our knowledge, the first machine-checked safety proof for a *production* Raft implementation, using a modern proof assistant without external libraries, and generated entirely by an automated LLM agent.

## 8 CONCLUSION

Formal verification has historically required deep expertise in proof assistants and months of dedicated human effort. This paper demonstrates that LLM-driven agents can eliminate that barrier.

Starting from zero, the Lean Squad agent—running as a scheduled GitHub Actions job with no human in the loop—autonomously selected 22 verification targets in a production Raft implementation, designed a six-layer proof architecture, wrote all specifications and proofs, and produced 443+ machine-checked theorems across 23 files in 34 runs, culminating in the end-to-end safety theorem `raftReachable_safe` with zero `sorry`. The human author's total involvement was: designing the reusable workflow, merging PRs, and writing two short comments. He required no Lean expertise, no formal-methods training, and invested near-zero hours of proof-engineering labour.

The key enablers are: (i) a phase-weighted task selection mechanism that gives the agent a well-defined next action at every stage of the project; (ii) persistent memory that accumulates proved lemmas, target status, and proof strategies across independent runs; and (iii) the dramatically improved reasoning capabilities of modern coding agents (Copilot CLI with Claude Opus 4.6 in this project).

The proof architecture is layered, honest, and extensible: each layer's residual assumptions are explicit, and the path to a fully concrete protocol proof is clear. The Lean Squad workflow is open-source and installs on any GitHub repository with two commands (§3.6), requiring only an LLM API token. We believe this shifts the economics of formal verification: the cost moves from scarce human proof-engineering expertise to commodity LLM compute, making machine-checked safety proofs a practical option for ordinary software teams.

Future work on Lean Squad includes reliably applying the Aeneas [18] toolchain to mechanically extract Lean models from safe Rust; generalising it to use a broader range of formal verification tools and methodology; and applying Lean Squad to other production codebases to test generality.

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